Scientific Computing
Week 1

We will cover two topics this week: Monday we analyse the approximations needed in order to make a scientific problem computable, and quantify the different errors this introduces. Wednesday we learn about linear systems and algorithms for solving them.

The times and locations are the same every week: please find them at the course description page.

Monday: Computational Models and Their Errors

Topics:	Intro to scientific computing Approximations introduced by computational models Quantifying and bounding errors Fixed- and floating-point numbers.
Read beforehand:	Chapter 1.1-1.3
Review Questions:	1.1-1.9
Exercises:	1.3,1.6

Wednesday: Solving Systems of Linear Equations

Topics:	Systems of linear equations as matrix equations Rank, kernel, image, and solution spaces Data-sensitivity, conditioning Algorithm 1: Gaussian elimination Algorithm 2: LU-factorization
Read beforehand:	Chapter 2.1-2.4.4
Review Questions:	2.1 – 2.5, 2.21 – 2.25, 2.28, 2.29, 2.62
Exercises:	2.4, 2.17